

ROHAN JUNEJA

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RESEARCH INTERESTS

I work at the intersection of VLSI design, computer architecture, and AI, developing hardware-software co-design solutions to address performance and efficiency bottlenecks in the post Dennard era. My research focuses on reconfigurable architectures and GPU-based accelerators for irregular and mixed-precision workloads driven by sparsity and quantization, key enablers of AI model compression. I design LLVM-based compiler toolchains to map compressed models efficiently onto custom accelerators, and validate these designs through real-world chip tapeouts. By advancing performance, energy efficiency, and reliability, my work aims to lower the carbon footprint of modern computing systems.

EDUCATION

- **National University of Singapore**
Ph.D. in Computer Science (CGPA: 4.58/5) *January 2021 - Present*
Advisors: [Prof. Li-Shiuan Peh](#), [Prof. Tulika Mitra](#)
Thesis: Bridging Model and Silicon: Co-Design for Efficient and Robust AI Inference
- **IIIT Delhi**
B.Tech in Electronics and Communications Engineering (CGPA: 8.41/10) *May 2014 - May 2018*
Advisors: [Prof. Lam Siew Kei](#), [Prof. Sujay Deb](#)
Thesis: Securing untrusted memories in embedded systems

PROFESSIONAL EXPERIENCE

- **IBM Research** Yorktown Heights, New York
PhD Research Intern *Nov 2025 - Jan 2026*
 - Designed and optimized collective communication primitives for multi-AIU systems to reduce latency and improve scalability.
- **National University of Singapore** Singapore
PhD Researcher *Jan 2021 - Present*
 - Designed low-power edge accelerators for transfer learning and transformer-style attention workloads, targeting high-throughput inference.
 - Co-designed PACE, a CGRA integrated with a RISC-V Rocket core, contributing to hardware design and LLVM-based compiler development; achieved 360 GOPS/W.
 - Validated PACE using FPGA-based pre-silicon emulation, supporting functional verification and performance optimization before fabrication.
 - Developed reconfigurable architectures with sparsity and multi-precision support to improve scalability and energy efficiency.
 - Designed hardware-aware quantization techniques for TPUs and GPUs to improve inference efficiency for LLMs and other AI models. Developed CUDA kernels using the CUTLASS library to exploit hardware characteristics and maximize throughput.
- **Advanced Micro Devices** Singapore
PhD Research Intern *May 2022 - July 2022*
 - Designed an accelerator for Ethereum's Beacon Chain using parallel processing of Tamper-Evident Plan (TEP) trees to improve transaction validation and execution.
 - Improved performance modeling and profiling techniques for blockchain workloads.
- **Renesas Electronics Corporation** Singapore
PhD Engineering Intern *Jan 2022 - April 2022*

- Developed deep learning acceleration support for Renesas' Dynamically Reconfigurable Processor (DRP), focusing on dynamic architecture reconfiguration.
- Engineered a smart gym application using DRP-based acceleration to improve computational efficiency and responsiveness.

• Qualcomm

CPU Design Engineer

Bengaluru, India

July 2018 - January 2021

- Worked as a CPU design engineer for Qualcomm Snapdragon Processors.
- Delivered multi-clock-domain and low-power UPF RTL for ARM Kryo cores in Snapdragon 765G, medium-tier, high-tier, and compute chips.
- Responsible for restructuring memory model RTL to support partial power gating.
- Worked on Power Manager IP, DCVS, ARM P-channel low-power modes, and boot RTL in Snapdragon CPUs.
- Wrote SystemVerilog assertions and closed code coverage and functional coverage.
- Used C, C++, Python, SystemVerilog, synthesis flows, and performance profiling for RTL development and optimization.

PUBLICATION RECORD

Conferences

1. Nexus Machine: An Active Message Inspired Reconfigurable Architecture for Irregular Workloads *MICRO 2025*
Rohan Juneja, Pranav Dangi, Thilini Kaushalya, Tulika Mitra, Li-Shiuan Peh
2. Building an Open CGRA Ecosystem for Agile Innovation *ICCAD 2025*
Rohan Juneja, Pranav Dangi, Thilini Kaushalya, Zhaoying Li, Dhananjaya Wijerathne, Li-Shiuan Peh, Tulika Mitra
3. HALO: Hardware-aware quantization with low critical-path-delay weights for LLM acceleration *AAAI 2026*
Rohan Juneja, Shivam Aggarwal, Safeen Huda, Tulika Mitra, Li-Shiuan Peh
4. Hardware-Aware Quantization for Efficient LLM Acceleration *MICRO 2025 [SRC Winner]*
Rohan Juneja, Safeen Huda, Tulika Mitra, Li-Shiuan Peh
5. EcoCGRA: Harnessing CGRA Programmability for Fault-Tolerant and Sustainable Acceleration *MICRO 2026 [Under Review]*
Rohan Juneja*, Pranav Dangi*, Lieven Eeckhout, Tulika Mitra
*Equal contribution
6. COMPOSE: Static Timing-driven Composable Reconfigurable Architecture *MICRO 2026 [Under Review]*
Rohan Juneja, Vishruti Ranjan, Rakshith Harish, Li-Shiuan Peh
7. EdgeWizard: A Motion-Resilient EEG Wearable with On-Device Processing Using Reconfigurable Fabric *TBioCAS 2026 [Under Review]*
Rohan Juneja*, Teck Lun Goh*, Vishruti Ranjan, Shivam Aggarwal, Zhaoying Li, Tulika Mitra, Li-Shiuan Peh
*Equal contribution
8. A Data-Driven Dynamic Execution Orchestration Architecture *ASPLOS 2026*
[Best Paper Honorable Mentions]
Pranav Dangi, Zhenyu Bai, **Rohan Juneja**, Zhaoying Li, Zhanglu Yan, Huiying Lan, Tulika Mitra
9. Enhancing CGRA Efficiency through Aligned Compute and Communication Provisioning *ASPLOS 2025*
Zhaoying Li, Pranav Dangi, Chenyang Yin, Thilini Kaushalya, **Rohan Juneja**, Cheng Tan, Zhenyu Bai, Tulika Mitra
10. ZeD: A Generalized Accelerator for Variably Sparse Matrix Computations in ML *PACT 2025*
Pranav Dangi, Zhenyu Bai, **Rohan Juneja**, Dhananjaya Wijerathne, Tulika Mitra
11. NOVA: NoC-based Vector Unit for Mapping Attention Layers on a CNN Accelerator *DATE 2024*
Mohit Upadhyay, **Rohan Juneja**, Weng-Fai Wong, Li-Shiuan Peh
12. FLEX: Introducing FLEXible Execution on CGRA with Spatio-Temporal Vector Dataflow *ICCAD 2023*
Thilini Kaushalya, Dan Wu, **Rohan Juneja**, Dhananjaya Wijerathne, Tulika Mitra, Li-Shiuan Peh

13. REACT: A Heterogeneous Reconfigurable Neural Network Accelerator with Software Configurable NoCs for Training and Inference on Wearables *DAC 2022*
Mohit Upadhyay, **Rohan Juneja**, Bo Wang, Jun Zhou, Weng-Fai Wong, Li-Shiuan Peh
14. Cache-Aware Dynamic Skewed Tree for Fast Memory Authentication *ASP-DAC 2021*
Saru Vig, **Rohan Juneja**, Siew Kei Lam
15. DISSECT: Dynamic Skew-and-Split Tree for Memory Authentication *DATE 2020*
Saru Vig, **Rohan Juneja**, Siew Kei Lam, Guiyuan Jian
16. Dynamic NoC Platform for Varied Application Needs *ISQED 2018*
Sidhartha Shankar, Hemanta K. Mondal, **Rohan Juneja**, Sri Harsha Gade, Sujay Deb

Journals

1. CTScan: A CGRA-based Platform for Emulation of Power Side-Channel Attacks on Edge CPUs *TRETS 2025*
Yaswanth Tavva, **Rohan Juneja**, Trevor E. Carlson, Li-Shiuan Peh
2. Framework for Fast Memory Authentication using Dynamically Skewed Integrity Tree *TVLSI 2019*
Saru Vig, **Rohan Juneja**, Guiyuan Jiang, Siew Kei Lam, Changhai Ou

Chip Tapeouts

1. PACE: A Scalable and Energy Efficient CGRA in a RISC-V SoC for Edge Computing Applications *HotChips 2024*
Vishnu Nambiar, Yi Sheng Chong, Thilini Kaushalya, Dhananjaya Wijerathne, Zhaoying Li, **Rohan Juneja**, Li-Shiuan Peh, Tulika Mitra, Anh Tuan Do
2. A 360 GOPS/W CGRA in a RISC-V SoC with Multi-Hop Routers and Idle-State Instructions for Edge Computing Applications *ISOC 2024*
Vishnu Nambiar, Yi Sheng Chong, Thilini Kaushalya, Dhananjaya Wijerathne, Zhaoying Li, **Rohan Juneja**, Li-Shiuan Peh, Tulika Mitra, Anh Tuan Do
3. HiVec: Scalable and Energy Efficient CGRA in a RISC-V SoC for Wearables *HotChips 2026 [Under Review]*
Rakshith Harish*, **Rohan Juneja***, Pranav Dangi, Zhenyu Bai, Vishnu Nambiar, Yi Sheng Chong, Bin Zhao, Vishruti Ranjan, Rahul Dutta, Li-Shiuan Peh, Tulika Mitra and Anh Tuan Do
*Equal contribution

Patents

1. A Reconfigurable Execution Unit for Collective Routing and Computation of Multiple Operations for Hardware Acceleration *Patent 10202402819X, 2025*
Zhaoying Li, **Rohan Juneja**, Pranav Dangi, Tulika Mitra

TEACHING EXPERIENCE

- **National University of Singapore (NUS)**
Teaching Assistant, Introduction to Operating Systems *January 2023 – April 2023*
 - Independently led weekly tutorial sessions for a cohort of 48 undergraduate students, facilitating an in-depth understanding of core operating systems concepts.
 - Provided detailed feedback on student assignments and exams, fostering continuous improvement and strengthening analytical skills.
- **National University of Singapore (NUS)**
Teaching Assistant, Introduction to Operating Systems *August 2022 – November 2022*
 - Delivered interactive tutorial sessions tailored to a class of 48 students, emphasizing problem-solving and practical application of OS principles.
 - Supported Prof. Weng-Fai Wong in streamlining assessments, including the design, evaluation, and grading of assignments and examinations.